

AI Smart City Management System

Introduction

Municipal administration is an essential part of urban development that provides civic services and infrastructure management for citizens. Managing citizen records, allocating complaints to field departments, tracking repair progress, and gathering public feedback manually can be time-consuming and may lead to delays and errors. The AI Smart City Management System is a web-based application designed to simplify municipal administration by automating civic complaint tracking and routing digitally. In this system, the administrator registers field officers, manages city infrastructure records, and monitors department analytics, while citizens can log in to report complaints, track issue timelines, and give feedback. Officers can also log in to access their assigned tasks and get instructions on repair procedures.

Abstract

The AI Smart City Management System is a web-based application developed using Python, Flask, and SQLite. The administrator can register field officers, manage departments, monitor city-wide KPIs, and allocate complaints. Citizens log in using secure credentials to submit civic complaints with pictures and descriptions, track progress in real-time, and submit feedback ratings. The system integrates the Google Gemini 2.5 Flash Multimodal AI Engine to automatically read submitted texts and images, categorize them into specific departments (such as Roads & Potholes, Sanitation & Waste, Water Supply), assess risk priority levels, and auto-route them to available officers. An interactive chatbot, CityBot, leverages database context to guide citizens with municipal information and help officers with step-by-step repair Standard Operating Procedures (SOPs).

Tools & Technologies

Frontend

- HTML5
- CSS3 (Vanilla CSS)
- JavaScript (ES6+)

Backend

- Python (v3.10+)
- Flask (v3.0.0+)
- Flask-Login (v0.6.3+)
- Flask-SQLAlchemy (v3.1.0+)
- Google GenAI SDK (Gemini 2.5 Flash API)

Database

- SQLite3

Development Tools

- Visual Studio Code (VS Code)
- Git & GitHub
- Dotenv (Environment Configuration)

Modules

Admin Authentication

- Admin Login
- Secure Session Management
- Role-Based Authorization (Admin Level)

Citizen Management

- Citizen Registration
- Profile Management (Ward, Address, Phone)
- In-App Notifications Log

Officer Management

- Register Officer Profiles
- Department Allocation (Sanitation, Roads, Electricity, etc.)
- Track Officer Workload (Active Assigned & Resolved counts)
- Officer Rating Monitoring

Complaint Management

- Submit Complaint (Title, Description, Category, Location, Coordinates, Image upload)
- Multimodal AI Classification (Gemini 2.5 Flash Image & Text processing)
- Local Keywords Fallback Rules Engine
- Timeline Tracking (Immutable log of updates per ticket)

Auto-Dispatch & Routing

- Category-to-Department Mapping
- Available Officer Filter
- Workload Balancing Dispatch (Routes to officer with the least active tickets)

AI Assistant (CityBot)

- Interactive AI Chat Interface
- Multi-Turn Conversation Memory (SQLite database-saved history)
- Role-Aware Prompts (Technical SOPs for Officers, Action links for Citizens, Summary statistics for Admins)
- Local Offline Response Backup

Feedback Management

- Submit Rating Feedback (1-5 Stars and Comments)
- View Complaint Feedback Logs (For Admins and Officers)

Dashboard

- Citizen Dashboard (Individual stats, recent complaints status, unread alert count)
- Officer Dashboard (Task queue sorted by priority, department ticket lists, performance metrics)
- Admin Dashboard (KPI counts, category breakdowns, monthly trends, top-performing officers list, department resolution rates)

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